

INFRASTRUKTUR SERVER INTERNAL

Otomatisasi CI/CD & Lingkungan Review Dinamis

Implementasi self-hosted GitLab, Traefik, dan manajemen konfigurasi terpusat.

Arsitektur Ekosistem DevOps



GitLab CI/CD

Pusat eksekusi pipeline.



Code-Server

Injeksi konfigurasi terpusat.



Harbor Registry

Distribusi custom image.



Portainer

Monitoring dan orkestrasi node.



Traefik Proxy

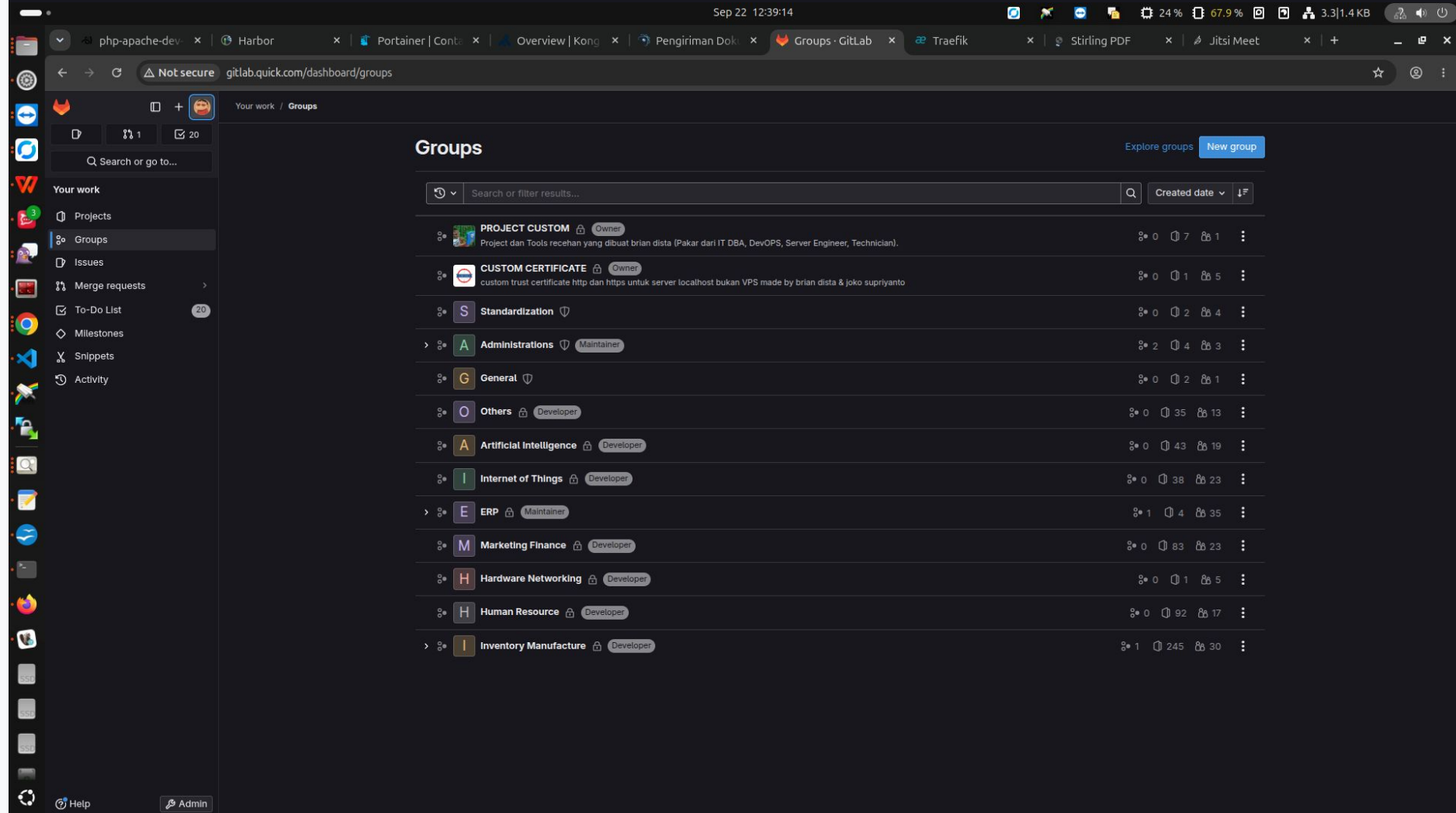
Manajemen routing dinamis.

MANAJEMEN KODE

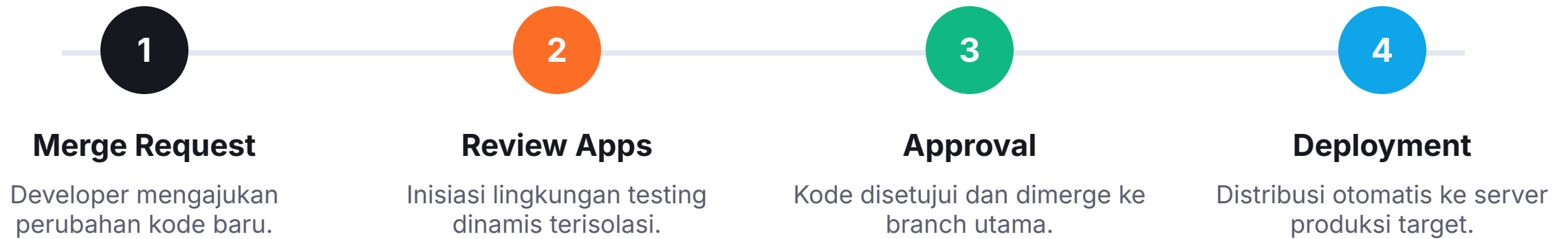
Struktur Repositori

Klasifikasi hierarkis berdasarkan grup divisi dan nama proyek.

- ✓ Akses terkontrol.
- ✓ Rapih dan terukur.



Alur Kerja Otomatisasi



```
1 services:
2   web:
3     image: reg.docker.quick.com/webdevops/php-apache-dev:5.6-0.0.1
4     container_name: $CI_PROJECT_ROOT_NAMESPACE.$CI_PROJECT_NAME.$CI_MERGE_REQUEST_ID.web
5     restart: unless-stopped
6     user: "$USER_GROUP_ID"
7     userns_mode: "host"
8     environment:
9       - WEB_PHP_TIMEOUT=5000
10      # - XDEBUG_MODE=profile,trace
11      # - XDEBUG_START_WITH_REQUEST=yes
12      # - XDEBUG_PROFILER_ENABLE=1
13      # - XDEBUG_PROFILER_OUTPUT_DIR=/xdebug
14      - CI_ENV=development
15      - php.session.save_path=/tmp/php-sessions
16     volumes:
17       - ./www:/app
18       - ./www/xdebug:/xdebug
19       - ./secrets:/secrets
20
21     command: >
22       sh -c "
23         curl -sS https://getcomposer.org/installer | php -- --2 --install-dir=/usr/local/bin --filename=composer &&
24         apache2-foreground
25       "
26
27     deploy:
28       resources:
29         limits:
30           memory: 1000M
31
32     labels:
33       - "traefik.enable=true"
34       - "traefik.http.routers.$CI_MERGE_REQUEST_ID-$CI_COMMIT_SHA.rule=Host('$CI_MERGE_REQUEST_ID.$CI_PROJECT_NAME.$CI_PROJECT_ROOT_NAMESPACE.staging.quick.com')"
35       - "traefik.http.routers.$CI_MERGE_REQUEST_ID-$CI_COMMIT_SHA.entrypoints=http"
36       - "traefik.http.routers.$CI_MERGE_REQUEST_ID-$CI_COMMIT_SHA.tls=false"
37
38     networks:
39       - traefik
40
41 networks:
42   traefik:
43     name: traefik proxy
44     external: true
```

INJEKSI DEPENDENSI

Konfigurasi Terpusat

Pengambilan template Docker Compose dan konfigurasi koneksi on-the-fly.

Bebas hardcoded.

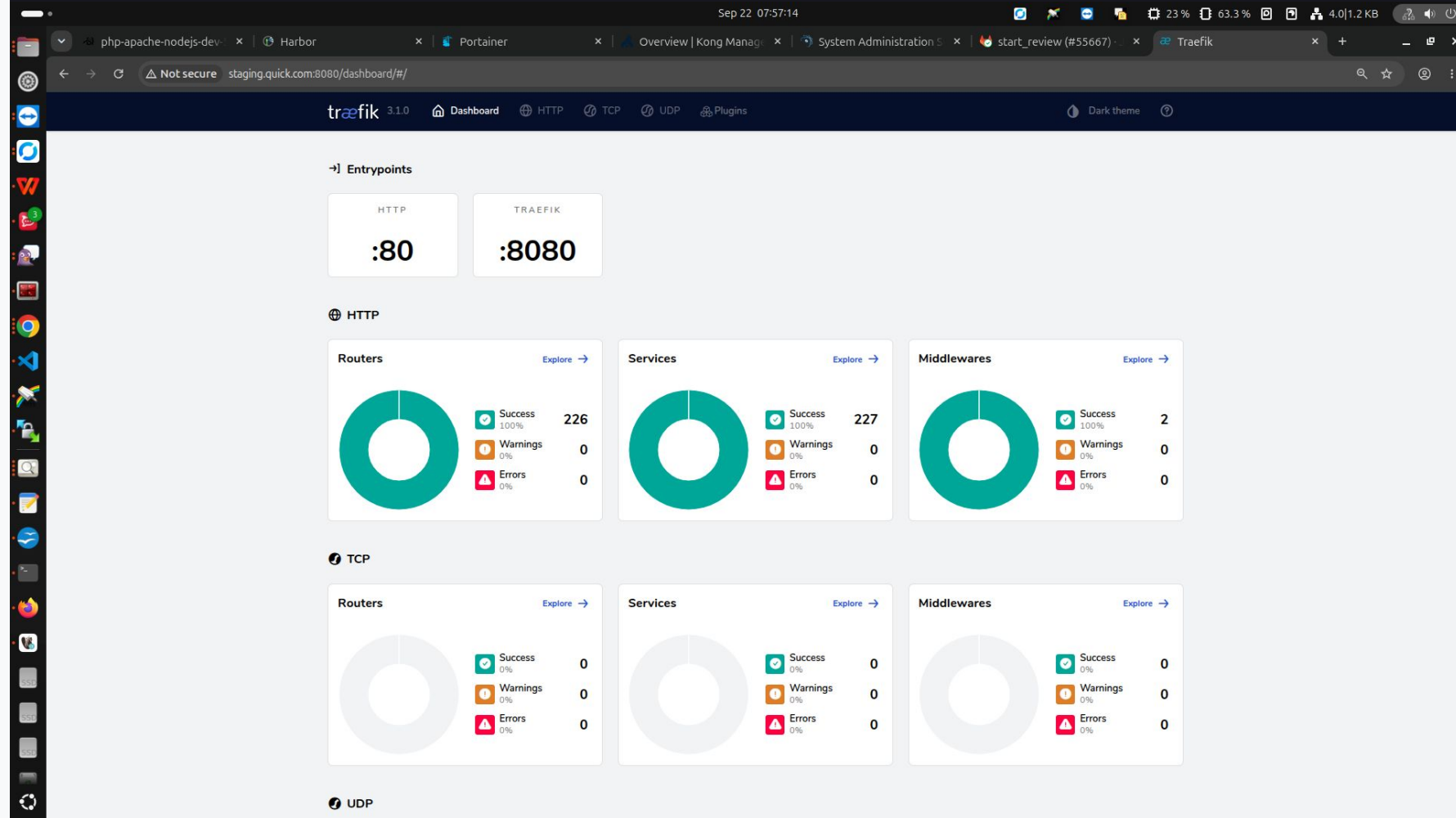
Setiap proyek mengambil file spesifik jalurnya via perintah curl otomatis.

NETWORK MANAGEMENT

Routing Dinamis Traefik

Mengelola puluhan lingkungan Review Apps secara paralel.

- 🛡️ Resolusi DNS otomatis.
- 🛡️ Anti tabrakan port antar kontainer.



Validasi Kode Instan

Sep 22 07:55:16

Walkthrough: Setup VS C x Harbor Portainer Viewing route | Kong Ma x System Administration S x start_review (#55667) x

Not secure gitlab.quick.com/erp/web/-/jobs/55667

ERP / Web / Jobs / #55667

Search visible log output

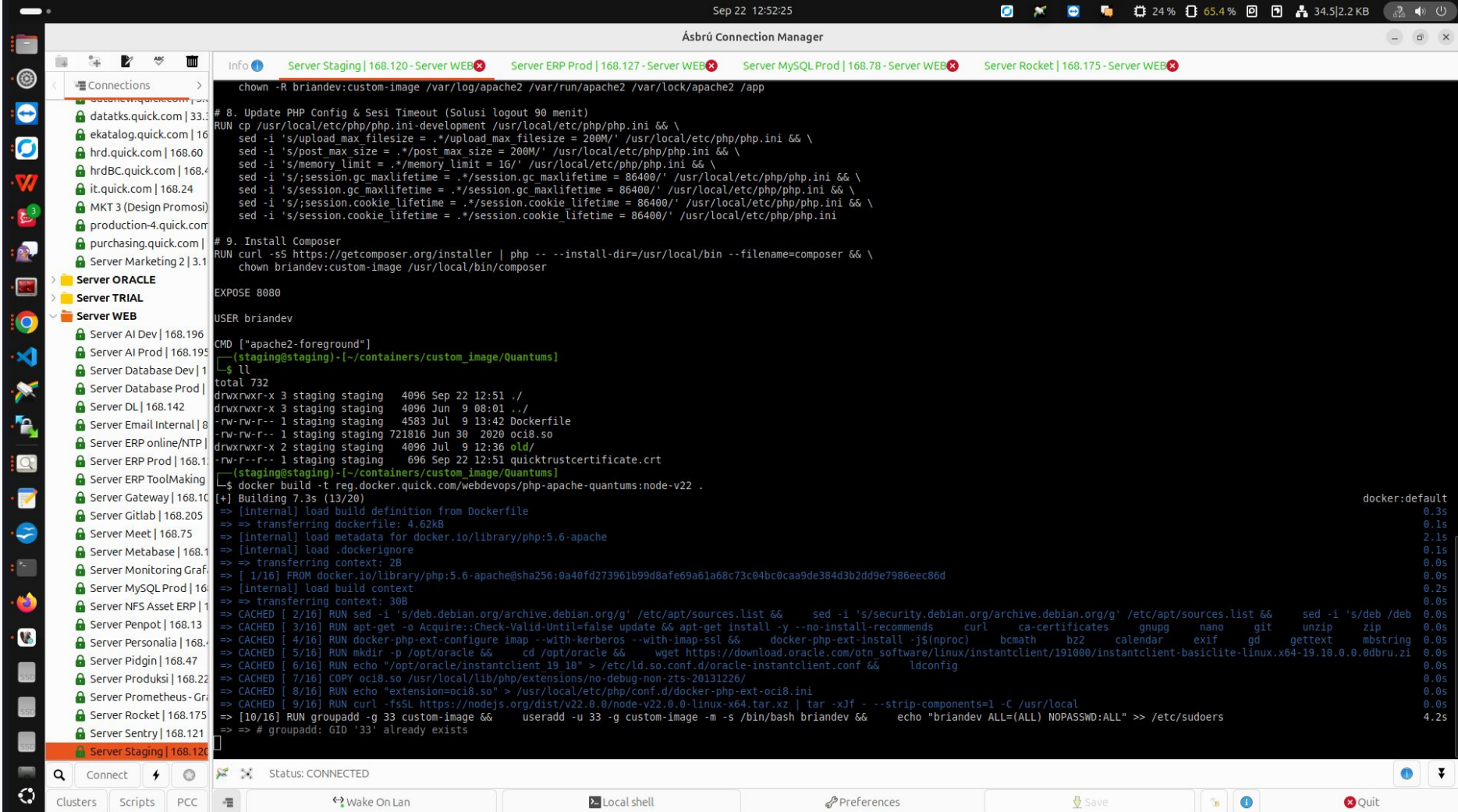
Duration: 3 minutes 24 seconds
Finished: 27 minutes ago
Queued: 1 second
Timeout: 1h (from project)
Runner: #15 (jVhQYzXQE) GitLab shell runner in staging server
Tags: staging-shell-runner
Commit 46c69ed0 in I6000
fix: Input aset opname image
Pipeline #24230 Blocked for I6000
with bugfix/T0828/of1-asset-opname
review_apps

Related jobs
stop_review
start_review

```
44 - Installing setasign/fpdf (v2.6.0): Downloading (connecting...) Downloading (0%) Downloading (1
0%) Downloading (15%) Downloading (25%) Downloading (40%) Downloading (50%) Downloading (6
5%) Downloading (75%) Downloading (80%) Downloading (98%) Downloading (100%) Downloading (1
45 - Installing setasign/fpdf (1.8.6): Downloading (connecting...) Downloading (0%) Downloading (5%) Do
wnloading (10%) Downloading (15%) Downloading (20%) Downloading (25%) Downloading (30%) Downloading
(35%) Downloading (40%) Downloading (45%) Downloading (50%) Downloading (55%) Downloading (6
0%) Downloading (65%) Downloading (70%) Downloading (75%) Downloading (80%) Downloading (8
5%) Downloading (90%) Downloading (95%) Downloading (100%)
46 - Installing clegginabox/pdf-merger (dev-master 7383e41): Cloning 7383e41bfc
47 - Installing psr/cache (1.0.1): Downloading (connecting...) Downloading (0%) Downloading (45%) Down
loading (100%)
48 - Installing firebase/php-jwt (v5.5.1): Downloading (connecting...) Downloading (0%) Downloading (6
0%) Downloading (80%) Downloading (100%) Downloading (1
49 - Installing google/auth (v1.19.1): Downloading (connecting...) Downloading (0%) Downloading (5%) Do
wnloading (10%) Downloading (40%) Downloading (45%) Downloading (50%) Downloading (55%) Downloading
(60%) Downloading (65%) Downloading (70%) Downloading (75%) Downloading (100%)
50 - Installing mykyawhtun/pdfmerger (dev-master a0e6fca): Cloning a0e6fcaf10
51 - Installing ramsey/uuid (3.x-dev ad596eb): Cloning ad596ebba
52 - Installing sentry/sentry (1.11.0): Downloading (connecting...) Downloading (100%)
53 - Installing vlucas/phpdotenv (4.3.x-dev 67a491d): Cloning 67a491df68
54 firebase/php-jwt suggests installing paragonie/sodium_compat (Support EdDSA (Ed25519) signatures when libsodium is not present)
55 google/auth suggests installing phpseclib/phpseclib (May be used in place of OpenSSL for signing strings or for token management. Please require version ^2.)
56 ramsey/uuid suggests installing ext-libsodium (Provides the PECL libsodium extension for use with the SodiumRandomGenerator)
57 ramsey/uuid suggests installing ext-uuid (Provides the PECL UUID extension for use with the PeclUuidTimeGenerator and PeclUuidRandomGenerator)
58 ramsey/uuid suggests installing moontoast/math (Provides support for converting UUID to 128-bit integer (in string form).)
59 ramsey/uuid suggests installing paragonie/random-lib (Provides RandomLib for use with the RandomLibAdapter)
60 ramsey/uuid suggests installing ramsey/uuid-console (A console application for generating UUIDs with ramsey/uuid)
61 ramsey/uuid suggests installing ramsey/uuid-doctrine (Allows the use of Ramsey\Uuid\Uuid as Doctrine field type.)
62 sentry/sentry suggests installing monolog/monolog (Automatically capture Monolog events as breadcrumbs)
63 Generating autoload files
64 8 packages you are using are looking for funding.
65 Use the 'composer fund' command to find out more!
66 $ if [ -f compose.yml ]; then # collapsed multi-line command
67 % Total % Received % Xferd Average Speed Time Time Time Current
68 Dload Upload Total Spent Left Speed
69 100 73685 100 73685 0 0 3182k 0 ----- 3270k
70 $ if [ -f compose.yml ]; then # collapsed multi-line command
71 % Total % Received % Xferd Average Speed Time Time Time Current
72 Dload Upload Total Spent Left Speed
73 100 43 100 43 0 0 6715 0 ----- 7166
74 $ if [ -f compose.yml ]; then # collapsed multi-line command
75 php-fpm:php-fpm: stopped
76 php-fpm:php-fpm: started
77 Job succeeded
```

Custom Docker Image

- ⚙️ Konfigurasi PHP & limitasi.
- 📁 Bypass limitasi image publik.



Sep 22 07:55:57

Walkthrough: Setup VS x Harbor Portainer Viewing route | Kong Ma x System Administration S x start_review (#55667) x +

reg.docker.quick.com/harbor/projects

Harbor Search Harbor... English Default brian_dista_setya_budi

Projects

Logs Administration Users Robot Accounts Registries Replications Distributions Labels Project Quotas Interrogation Services Clean Up Job Service Dashboard Configuration

Projects

+ NEW PROJECT ACTION

All Projects

☐	Project Name	Access Level	Role	Type	Repositories Count	Creation Time
☐	android	Public	Project Admin	Project	1	10/4/24, 1:29 PM
☐	apps	Public	Project Admin	Project	2	10/18/24, 1:03 PM
☐	custom	Public	Project Admin	Project	1	6/17/26, 11:04 AM
☐	development	Public	Project Admin	Project	0	6/16/25, 2:04 PM
☐	docker-registry	Public	Project Admin	Proxy Cache	11	9/10/24, 10:00 AM
☐	ghcr-registry	Public	Project Admin	Proxy Cache	2	10/3/24, 7:57 AM
☐	library	Public	Project Admin	Project	0	9/4/24, 3:38 PM
☐	oracle	Public	Project Admin	Project	2	11/18/24, 2:59 PM
☐	quantums	Public	Project Admin	Project	0	5/19/26, 3:50 PM
☐	rocket-chat	Public	Project Admin	Project	2	9/19/24, 1:50 PM
☐	tools	Public	Project Admin	Project	1	10/23/24, 8:51 AM
☐	web	Public	Project Admin	Project	1	10/9/24, 8:51 AM
☐	webdevops	Public	Project Admin	Project	5	9/5/24, 9:44 AM

Page size 15 1 - 13 of 13 items

LIGHT Harbor API V2.0

EVENT LOG

Projects

Repositories

Quota used

Private 0 Public 13 Total 13

Private 0 Public 28 Total 28

25.77 GiB

IMAGE DISTRIBUTION

Private Harbor Registry

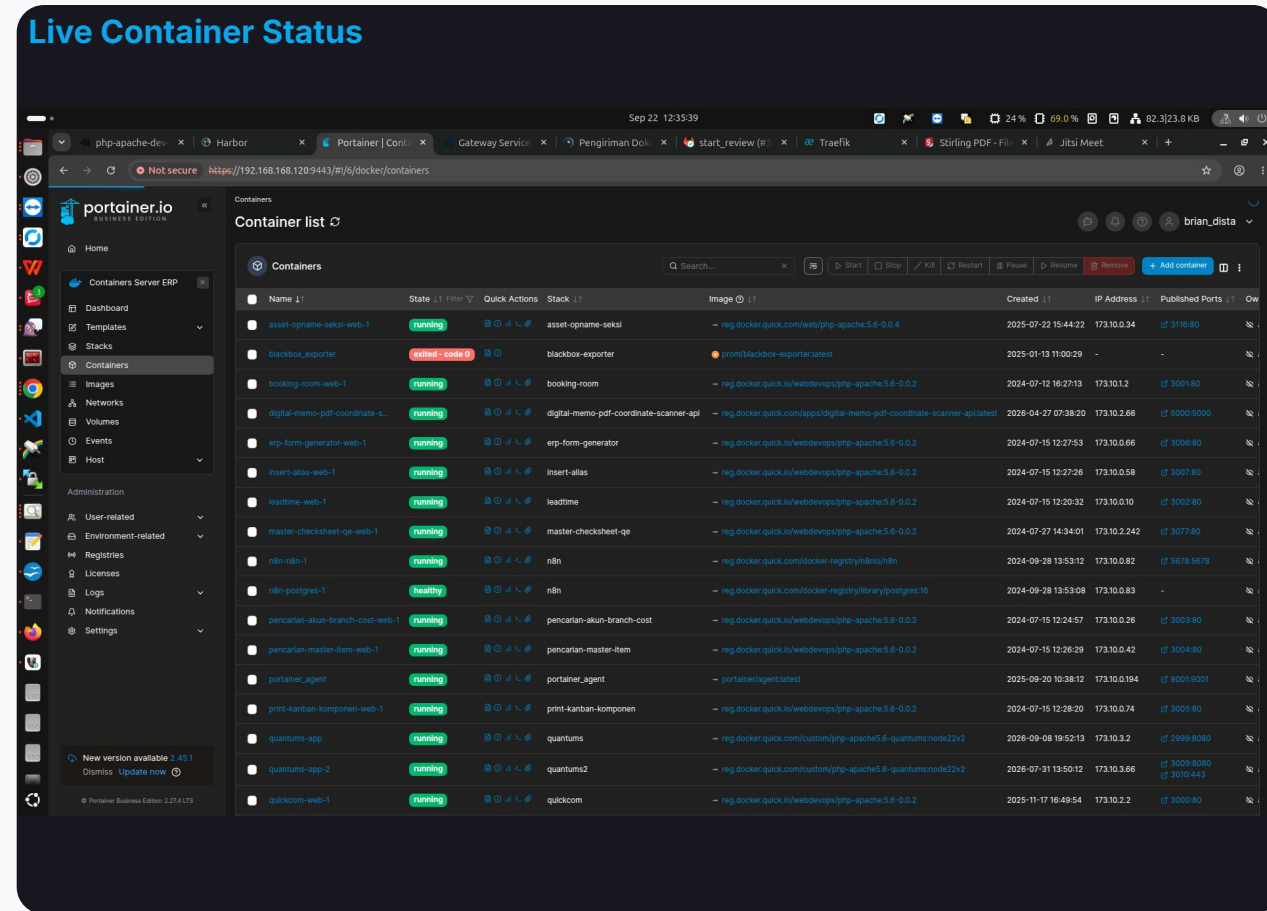
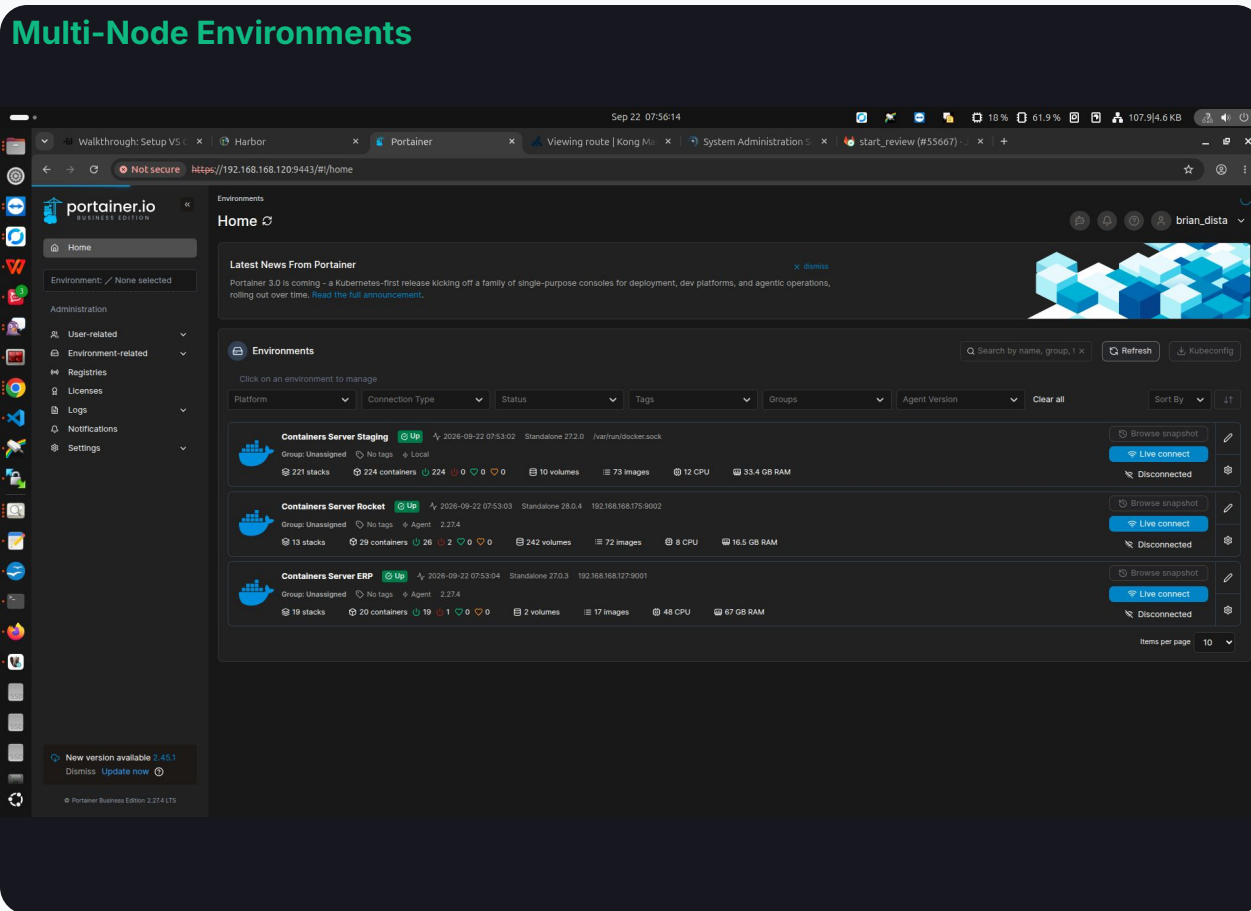
Pusat penyimpanan aman untuk mendistribusikan custom image ke seluruh environment.

Self-hosted mandiri.

Manajemen akses terkontrol untuk setiap proyek internal.

Sentralisasi Orkestrasi Portainer

Monitoring dan manajemen seluruh node dan kontainer dari satu dashboard.



Strategi Deployment

Target Spesifik

Penggunaan tags pipeline untuk menunjuk server produksi yang tepat.

Resolusi Konflik

Reset sinkronisasi dan pull otomatis dari branch utama.

Manajemen Paket

Eksekusi Composer instalasi on-the-fly jika terdeteksi perubahan.



Sistem Siap Beroperasi

Lingkungan pengembangan yang lebih efisien, terisolasi, dan terotomatisasi penuh.

